Q-SOTA: Enabling Post-Quantum OTA on Resource-Constrained ECUs*

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Abstract

With the proliferation of Software-Defined Vehicles (SDVs), Over-the-Air (OTA) updates have become a core mechanism for vehicle security. At the same time, the emergence of quantum computing threatens conventional public-key cryptographic systems, necessitating the transition to Post-Quantum Cryptography (PQC) in automotive OTA systems [1]. However, low-performance processor-based ECUs face significant limitations in memory and computational resources required for PQC signature verification [2]. To address this issue, this study proposes Q-SOTA (Quantum-Secure OTA). Q-SOTA adopts an architecture in which a high-performance Gateway performs PQC signature verification on behalf of ECUs and converts the verification result into a lightweight HMAC Token for delivery to the ECU. This design enables resource-constrained ECUs to achieve quantum-resistant security without directly executing PQC signature verification. In addition, inter-ECU version dependencies are validated through a Gateway-centric Version Graph verification structure (VG Summary), thereby structurally preventing version inconsistencies that may arise during partial updates. Comparative experiments with an ECDSA-based approach demonstrate that the performance overhead introduced by PQC transition remains within a practical range. Experiments were conducted using an STM32MP157D-DK1 (Cortex-A7) as the ECU and an NVIDIA Jetson TX2 (Denver 64-bit + A57 Quad-Core) as the Central Gateway. Five PQC algorithms—Falcon-512, Falcon-1024, and ML-DSA-44/65/87—were evaluated against an ECDSA (P-256) baseline. The Gateway-ECU security processing time of Q-SOTA (10.53–11.29 ms) achieved up to a 58% reduction compared to the ECDSA baseline (25.02 ms). This result shows that by offloading PQC computational overhead to the Gateway, the ECU only needs to perform lightweight HMAC Token verification, enabling even lower processing latency than direct ECDSA verification. Furthermore, the Cloud-Gateway signature transmission time differed by at most 1.9 ms from the ECDSA baseline, demonstrating that the network overhead caused by increased PQC signature sizes can be effectively absorbed at the architectural level. Nevertheless, this study assumes the Gateway to be a trusted entity, and countermeasures against Gateway compromise remain as future work.

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